HR ANALYTICS DASHBOARD

Power BI Project | GitHub README

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1. Project Overview

The HR Analytics Dashboard is a data analysis and visualization project developed using Microsoft Power BI. The purpose of this project is to help organizations understand their workforce by analyzing employee information such as employee details, salary distribution, gender, educational qualifications, joining dates, and leave balances.

An Excel dataset containing employee information was used as the data source. The data was prepared and transformed in Power Query and presented through interactive charts, graphs, and reports in Power BI. The dashboard provides an overview of workforce composition and helps users explore employee details and compare different categories.

2. Project Objectives

- Explore and understand employee data.
- Clean and organize raw data using Power Query.
- Analyze employee demographics and salary information.
- Study gender distribution across the organization.
- Understand employee joining trends and experience.
- Analyze employee leave balances.
- Create interactive reports that make HR data easier to understand.
- Present insights that can support HR decision-making.

3. Dataset Description

The project uses an Excel dataset containing the following employee-related fields:

- Employee Name – names of employees in the organization.
- Employee ID – a unique identification number for each employee.
- Gender – information about workforce gender distribution.
- Salary – salary details used to understand compensation patterns.
- Educational Qualification – employees' educational backgrounds and qualifications.
- Date of Joining – the date each employee joined the organization.
- Leave Balance – the number of available leave days for each employee.

The dataset was used to understand workforce composition and identify patterns in employee-related information.

4. Tools and Technologies

- Microsoft Power BI – building the interactive dashboard and visual reports.

- Microsoft Excel – primary source of employee data.
- Power Query – data cleaning and transformation.
- DAX (Data Analysis Expressions) – creating calculated columns and measures for analysis.

5. Project Workflow

1. Data exploration: examined the Excel dataset, its structure, important columns, and employee information.
2. Data cleaning and transformation: used Power Query to clean the data, address formatting issues, remove unnecessary information, and prepare it for analysis.
3. Data analysis: analyzed employee details, salary information, gender distribution, educational qualifications, joining dates, and leave balances.
4. Visualization: created charts, graphs, and visual elements to present HR information clearly.
5. Dashboard development: designed an interactive Power BI dashboard with filters to explore employee data by different categories.
6. Employee insights: included employee-level information such as job titles, age groups, and joining dates.

6. Dashboard Highlights

- Gender Analysis – displays the distribution of male and female employees.
- Job Title Analysis – presents employees across roles, including Production Operator and Research Analyst.
- Age Group Analysis – highlights employees in different age groups, including the 20–25 age group.
- Employee Details – enables exploration of individual employee information.
- Date of Joining – displays selected employees' joining dates and helps review joining patterns.
- Salary Analysis – supports exploration of salary information and compensation distribution.
- Leave Balance Analysis – provides an overview of available leave balances across employees.

7. Dashboard Design

The dashboard follows a clean, minimal, monochromatic design. A consistent colour theme helps keep the reports professional and easy to understand. The visual layout focuses on presenting important information clearly, without unnecessary distractions.

8. Key Learnings

Through this project, I gained practical experience working with employee datasets, cleaning and transforming data, creating DAX calculations, designing interactive dashboards, and presenting business insights using Power BI. The project also helped me understand how data analytics can support HR teams in monitoring workforce information and making informed business decisions.

9. Conclusion

The HR Analytics Dashboard demonstrates how raw employee data can be transformed into useful information through data analysis and visualization. It provides a straightforward way to explore workforce details, understand employee distribution, and monitor important HR metrics. This project strengthened my practical knowledge of Power BI, Excel, Power Query, and DAX, while highlighting the role of data analytics in Human Resource Management.

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Project: HR Analytics Dashboard using Power BI